

Paul Ghanem

ghanem.p@northeastern.edu | +1 (240) 302-7407 | 168 W sixth st, Apt 2, Boston, MA
github: github.com/paulghanem | website: paulghanem.github.io | scholar: scholar.google.com/citations?user=IOZaXkMAAAAJ

EDUCATION

Northeastern University <i>Ph.D. in Electrical Engineering</i> 2020 – 2025	Boston, MA
University of Maryland, College Park — A. James Clark School of Engineering <i>M.S. in Robotics</i> 2017 – 2019	College Park, MD
Lebanese American University <i>B.E. in Electrical Engineering</i> 2011 – 2016	Lebanon

SELECTED PUBLICATIONS

- **P. Ghanem**, T. Imbiriba, D. Erdogmus. "How Hard Should You Think? Self-Consistent Online Reinforcement Learning Meets Bayesian Latent Thought Optimization for Goodhart Aware Test-Time Reasoning." **Submitted (ICLR)**, 2026.
- **P. Ghanem**, O. Howell, D. Erdogmus. "Recursive Deep Inverse Reinforcement Learning." **Transactions on Machine Learning Research (TMLR)**, 2026.
- **P. Ghanem**, A. Demirkaya, A. Ramezani, Z. Danziger, D. Erdogmus. "Learning Physics Informed Neural ODEs With Partial Measurements." **AAAI**, 2025.
- **P. Ghanem**, D. Erdogmus, A. Ramezani. "Efficient Modeling of Morphing Wing Flight Using Neural Networks and Cubature Rules." **IEEE CDC**, 2023.
- **P. Ghanem**, A. Wolek, D. A. Paley. "Parallel and Circular Formation Control of a School of Robotic Fish." **ACC**, 2020.
- D. A. Paley, A. A. Thompson, A. Wolek, **P. Ghanem**. "Planar Formation Control of a School of Robotic Fish: Theory and Experiments." **Frontiers in Control Engineering**, 2021.
- S. S. Saab, **P. Ghanem**. "A Multivariable Stochastic Tracking Controller for Robot Manipulators Without Joint Velocities." **IEEE Trans. Automatic Control**, 2018.
- E. Sihite, **P. Ghanem**, A. Salagame, A. Ramezani. "Unsteady Aerodynamic Modeling of Aerobat Using Lifting Line Theory and Wagner's Function." **IEEE/RSJ IROS**, 2022.
- E. Sihite, B. Mottis, **P. Ghanem**, A. Ramezani, M. Gharib. "Efficient Path Planning and Tracking for Multi-Modal Legged-Aerial Locomotion Using Integrated Probabilistic Road Maps (PRM) and Reference Governors (RG)." **IEEE CDC**, 2022.
- B. Gupta, A. Salagame, **P. Ghanem**, E. Sihite, A. Ramezani. "Hovering Control of Flapping Wings in Tandem with Multi-Rotors." **IEEE/RSJ IROS**, 2023.

WORK EXPERIENCE

2026-Present Postdoctoral Researcher, LLM Reasoning & Post Training (LLM, Pytorch, HuggingFace) • Developed a Bayesian post-training framework using Ensemble Kalman Filtering to optimize latent thought embeddings in frozen LLMs. • Proposed a self-supervised online Reinforcement Learning method that learns per-problem optimization pressure of the Ensemble using self-consistency as a label-free reward, addressing Goodhart's Law without ground-truth labels, improving reasoning accuracy over state-of-the-art baselines across Llama-3.1-8B, Llama-3.1-70B, and Qwen-2.5-14B by up to 15%. Submitted results to ICLR.	University of Massachusetts, Boston, USA
2024 Research Intern, Audio ML & Active Noise Prediction/Cancellation (Python, TensorFlow, CUDA) • Trained a multimodal deep residual neural network for personalized noise prediction and cancellation. Through experimental evaluation, Identified that cancellation degrades when the driver's head position at inference differs from training, requiring head-tracking hardware. • Developed and trained Kalman filter-integrated Transformer that fuses roof microphone input filtered through the KF to remove noisy feedback, improving cancellation by 3–5 dB in high-frequency bandwidths. • Processed multi-gigabyte multimodal time-series data from vehicle-mounted acoustic and vibration sensors and collaborated with team members to deploy ML pipelines for cabin noise prediction.	Bose Corporation, Framingham, USA
2020-2025 Research Assistant, Inverse RL & Motion Planning/Prediction (MPC, IRL, RL, JAX, CUDA, PyTorch) • Developed end-to-end planning and trajectory forecasting algorithms for autonomous robotics that reason online under uncertainty. • Designed an inverse reinforcement learning algorithm, by merging Maximum Entropy Inverse Reinforcement Learning with MPPI, to learn moving radar's cost functions and policies. • Developed and validated inverse RL algorithms in OpenAI gym simulation environment (OpenAI Gym) for trajectory and end-to-end policy learning improving sample efficiency by ~50% and reward learning by ~30% over state-of-the-art inverse RL baselines and published the results in TMLR.	Northeastern University, Boston, USA
2020-2025 Research Assistant, Biomedical ML & Multimodal Data Integration (TensorFlow, PyTorch, RNN) • Developed a recursive ML method for training Neural Ordinary Differential Equations (Neural ODEs) from partially observed, multimodal biomedical data, recovering latent biological signals with 99% accuracy where existing methods fail to converge. Published in AAAI 2025. • Trained the Neural ODE model to recover hidden peripheral neuromodulation dynamics driving lower urinary tract (LUT) function from experimental animal data, supporting ML-driven computational approaches to treat bladder dysfunction.	Northeastern University, Boston, USA

- Designed a parallelized multi-subject deep learning framework that jointly trains across heterogeneous animal datasets to learn shared neuromodulation dynamics, addressing cross-subject variability, a core challenge in translational biomedical research.

2022-2023

Northeastern University, Boston, USA

Research Assistant, Robotic Perception & Sensor Fusion (OpenCV, Python, Modern C++17, ROS, Linux, Git, CNN)

- Deployed a multi-sensor perception system on the Hello Robot Stretch platform, integrating an RGB camera for face recognition and person re-ID with a lidar for leg tracking, fused through a unified Kalman filter framework in ROS for a Verizon-funded aging-in-place project.
- Validated the full pipeline in real-world experiments on the physical Stretch robot and published findings in PETRA.

2020-2025

Northeastern University, Boston, USA

Research Assistant, Aerial Robotics & Physics-Informed ML (MATLAB, Python, TensorFlow, Real-time Control)

- Designed a physics-informed neural network hybrid model for fluidic-structure interactions on a flapping-wing robot and developed an online training method using Kalman filters (EKF, CKF) to learn its unknown aerodynamic forces. Published in **CDC**
- Modeled aerodynamic forces using unsteady lifting line theory and Wagner functions, validated against experimental data on the physical aerobot platform. Published in **IROS**
- Assisted in hardware integration and real-time flight experiments on a robotic bat with an actuated spherical guard, contributing to observer-based controller design validated on the physical prototype. Published in **IROS**.

2021-2022

Northeastern University, Boston, USA

Teaching Assistant in Machine Learning (EECE 5644) (MATLAB, Python)

- Designed assignments for Introduction to Machine Learning (EECE 5644) covering Bayesian classification, GMMs, MLE/MAP estimation, EM, PCA, SVM, and gradient descent.
- Taught three lectures introducing students to training neural networks with Kalman filters and second-order methods.

2019-2020

Airgility.co, District of Columbia, USA

Controls Software Engineer, Autonomous Drones (C, MATLAB, Linux, Real-time Control)

- Assembled and integrated flight hardware, installed servo motors, wired actuators and a 9-DOF IMU to a lightweight embedded flight controller (ELKA board), and brought up a custom bi-copter UAV for a DHS-funded program.
- Implemented a Madgwick attitude estimator in C on the ELKA board, fusing 9-DOF IMU data to estimate yaw, pitch, and roll in real time for closed-loop flight control.
- Designed and deployed a PID attitude controller in MATLAB and C on embedded hardware (ELKA board), achieving stable hover with 1–2° attitude tracking error.

2018-2019

University of Maryland, College Park, USA

Research Assistant, Multi-Agent Control and Planning (C, MATLAB, Real-time Control)

- Developed decentralized multi-agent control for N underwater fish robots with oscillatory dynamics and nonholonomic constraints under limited communication, using Lyapunov stability and graph theory to guarantee parallel and circular formation convergence. Published in **ACC**.
- Implemented the decentralized controller in C on Arduino with onboard IMU sensing across a swarm of 4 physical fish robots, tuned and calibrated the hardware-software interface, and experimentally validated parallel and circular formations, achieving 3° heading tracking error under communication constraints. Published in **Frontiers in Control Engineering**.

2016-2017

Lebanese American University, Lebanon

Research Assistant, Robotic Manipulation & Bayesian Control (MATLAB, Real-Time Simulink, Optimal Control, Real-time Control)

- Wired and configured the real-time communication interface between a host PC (xPC target) and the Barrett 7-DOF robotic arm via CAN bus, setting up the CAN protocol and establishing reliable low-latency data exchange for closed-loop control at up to 1250 Hz.
- Developed a Bayesian adaptive PD controller that eliminates the need for velocity sensors by substituting fictitious velocity measurements and inflating their associated Bayesian controller covariances to account for the absence of true velocity data.
- Designed and executed all experiments on the physical 7-DOF manipulator in Real-Time Simulink, outperforming sliding mode control and standard PD baselines by 50% in joint tracking RMSE and 80% in torque vibration reduction, with performance gains scaling with sensor/controller sampling rates (500–1250 Hz), a property absent in competing methods. Published in IEEE **TAC**.

TECHNICAL SKILLS

Hardware & Experimental Platforms: Barrett WAM 7-DOF arm, Hello Robot Stretch, custom bi-copter UAV (ELKA board), underwater fish robots, vehicle-mounted acoustic/vibration sensor arrays, 9-DOF IMU, CAN bus

Languages & Frameworks: Python, Modern C++17, CUDA, MATLAB

ML & Scientific: PyTorch, JAX, TensorFlow, NumPy

Robotics & Systems: ROS, OpenCV, Simulink Real-Time, Linux

LLM & Inference: HuggingFace Transformers, vLLM, multi-GPU inference, model evaluation

SELECTED MEDIA COVERAGE

- **National Geographic**, "'Snakes' on the Moon? These Helpers Could Soon Join Our Lunar Mission," 2022
- **National Geographic**, "Bats Have Disease-Defying Superpowers, what if we could copy them? " 2024
- **Popular Science**, "Bats' Weird Wings Inspired This Drone," 2024
- **WIRED**, "Rovers Are So Yesterday. It's Time to Send a Snakebot to Space," 2023